ENGDATA Notes

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May 22, 2026

Contents

Type of Data

Study of the population is a **census**. Study of a sample of a population is a **survey**.

Data can either be numerical or categorical. Just because the data points are numbers, it doesn't mean they are numerical (e.g. student ID numbers, years).

The **Delphi technique** is when you send rounds of questionnaires to a panel of experts to reach a consensus.

Types of Probability Sampling

1. Simple Random Sampling: Every member has an equal chance of being selected (e.g. lottery, random number generator)
2. Cluster Sampling: The population is divided by clusters, which is often geographical. And entire clusters are randomly selected.
3. Stratified Sampling: The population is divided into subgroups (strata) based on shared characteristics and random samples are taken from each strata.
4. Systematic Sampling: Every n-th member of a group is selected.

The point of probability sampling is that each individual has a known, non-zero chance of being selected; minimizing bias and allowing for generalization.

Types of Non-Probability Sampling

1. Convenience Sampling: Individuals are selected for their availability and accessibility.
2. Voluntary Respone Sampling: Individuals select themselves.
3. Purposive/Judgmental Sampling: The researcher uses their expertise to select a sample that is most useful to the purpose of the research.
4. Snowball Sampling: Participants refer other participants.
5. Quota Sampling: Selecting a specific number of individuals from different subgroups.

These methods are faster and easier but risks higher bias. Most often used in exploratory research.

Types of Comparisons

1. Component comparison: "How does a part relate to the whole?" (pie chart)
2. Correlation comparison: "Do two variables move together?" (scatter plot, bubble chart)
3. Frequency comparison: "How many items fall into each range?" (histogram, bell curve)
4. Item comparison: "How do things rank against each other?" (bar chart, column chart)
5. Time series: "How does something change over time?" (line chart, area chart)

Probability

Defining Sample Space

Listing/Roster method: enumerating all elements of the sample space.

Property/Set-Builder method: Choosing a property or characteristic common to all sample points and then using this common characteristic to define the sample space.

Tree Diagram: Uses a tree data structure to express sequential order of events of an experiment in chronological order.

An **event** is a subset of a sample space.

A **sample point** is a specific outcome from the **sample space**.

A **simple event** is an event which only has one outcome.

A **compound event** is an event which consists of more than one outcome and may be decomposed into simple events.

Operations on Events

Union: A U B, set of all events in A or B.

Intersection: A $\cap$ B. Set of all events in A and B

Complement: A' represents the set of all events not found in A.

S represents the sample space. So S' represents all events not in the sample space.

A & B = NULL is **mutually exclusive** or **disjointed**.

Counting Principles and Techniques

Counting Theory

A set of techniques used for identifying the total number of possibilities of a statistical experiment without enumerating the elements of the sample space.

Multiplication Rule:

$n_T = n_1 \times n_2 \times ... \times n_k$

Where n is an outcome.

Addition Rule: If k outcomes are mutually exclusive, and outcome 1 can be performed in m_1 ways, outcome 2 can be performed in m_2 ways, and so on for k outcomes, then the total number of outcomes is given by:

$m_T = m_1 + m_2 + ... + m_k$

Simple Permutation

There is an ordered arrangement of **distinct objects**.

The formula:

$$nPr = \frac{n!}{(n-r)!}$$

Where n is the total number of objects and r is how many are being selected from n.

Simple Combination

An arrangement of distinct objects without regard to order.

$$nCr = \frac{n!}{(n-r)! * r!}$$

Special types of permutation

6.4.1 Circular Permutation

Arrangement of objects in a circle.

The formula is:

$$Pnc = (n-1)!$$

6.4.2 Arrangement of Similar Objects

To find the number of permutations of n "objects" all taken at the same time.

- n_1 are objects of type 1.
- n_2 are objects of type 2.
- n_k are objects of type k.

- n represents the total objects.

$$\sum_{i=1}^{k} n_i = n$$

This translates to:

$$\frac{n!}{n_1!n_2!n_3!...n_k!}$$

6.4.3 Partitioning of n objects into cells

To find the number of permutations of n "objects" all taken at the same time.

- n_1 are the objects placed in cell 1.
- n_2 are the objects placed in cell 2.
- n_k are the objects placed in cell k.

Very similar to the previous type of permutation.

$$\sum_{i=1}^{k} n_i = n$$

This translates to:

$$\frac{n!}{n_1!n_2!n_3!...n_k!}$$

Axioms of Probability

Axiom 1: 0 <= P(A) <= 1.

Axiom 2: P(A) = 1 if A is a certain or sure event.

Axiom 3: P(A) = 0 if A is an impossible event.

Axiom 4: If the probability of events in the sample space is added, the sum is always 1.

$$\sum_{i=1}^{n} P(E) = 1$$

Ways of measuring probability

Classical Concept

The probabilities of events can be determined without gathering data.

$$P(S) = \frac{n(A)}{N}$$

Where *n(A)* is the subset of the sample space and N is the total sample space.

Relative Frequency

Uses historical data or survey results to determine probabilities.

$$P(A) = \frac{f_A}{f_T}$$

Where f_A is the frequency of occurance of event A and f_T is the total number of observations.

Subjective Assessment

An expert in the field provides an educated guess of the probability.

Probability Rules

Union (A or B)

$$P(AUB) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive: $P(AUB) = P(A) + P(B)$

Intersection (A and B)

$$P(A \cap B) = P(A) \times P(B|A)$$

If A and B are independent: $P(A \cap B) = P(A) \times P(B)$

Complement of A

$$P(A') = 1 - P(A)$$

Probability of B given A

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

1. Find the probablity of A and B happening.
2. Divide the probability of both events happening by the probability of event A

If A and B are independent:

$$\begin{aligned} P(B|A) &= \frac{P(A \cap B)}{P(A)} \\ P(B) &= \frac{P(A \cap B)}{P(A)} \\ P(B) \cdot P(A) &= P(A \cap B) \\ P(A \cap B) &= P(A) \cdot P(B) \end{aligned}$$

Venn Diagram

The entire sample space can be expressed as:

$$P(SampleSpace) = P(A \cap B') + P(A' \cap B) + P(A \cap B) + P(A' \cap B')$$

Contingency Table

A tool used to facilitate the calculation of probabilities involving two events.

	B	B'	Total
A	$P(A \cap B)$	$P(A \cap B')$	$P(A)$
A'	$P(A' \cap B)$	$P(A' \cap B')$	$P(A')$
Total	$P(B)$	$P(B')$	1

Conditional Probability

The probability that an event will occur given that some other event has already occured.

$$P(A|B) = \frac{P(A)}{P(B)}$$

Binomial Probability

Bayes Theorem

Unlike usualky where P(effect|cause) is calculated, Bayesian reasoning tries to find P(cause|effect).

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$